Higher National Diploma in Engineering (Electrical)

First Year, 1St Semester Examination — 2021

EE 1107 Electrical Principals A

Question 1

a. Briefly explain the following terms used in electrical circuits:

   i. **Active networks**.
   ii. **Passive networks**.
   iii. **Linear circuits**. [06 marks]

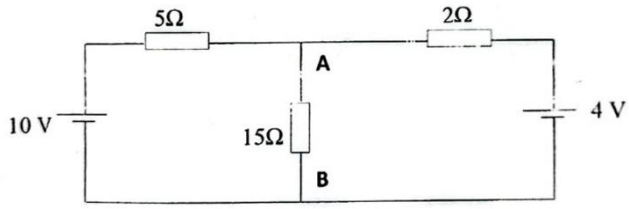

Figure Q1

b. Figure Q1 shows a linear circuit. Find the current through 15Ω resistor shown in the circuit by using the following methods:

   i. Kirchhoff's laws.
   ii. Superposition Theorem.[10 marks]

c. Obtain **Thevenin equivalent circuit** across points A and B as shown in Figure Q1.[04 marks]

Question 2

(a). A parallel plate capacitor is constructed using two circular plates of area A. The gap (separation) between the plates is d, and the gap is filled with a dielectric of relative permeability G. Derive an equation for the capacitance of the capacitor.[02 marks]

(b). It is needed to construct a parallel plate capacitor, where **Bakelite** is used as the dielectric filling. The capacitance requirement is $0.05\mu\mathrm{F}$ and the maximum value of steady working potential is $1.2\ \mathrm{kV}$. Allowing a safe value of field stress of $25\ \mathrm{MV/m}$, find the followings: (i). the thickness of the Bakelite required. (ii) the area of the plate required if the relative permittivity of Bakelite is 5. (iii) the maximum energy stored by the capacitor. (iv) the average power developed if this energy is dissipated in a time of $20\mu\mathrm{s}$.[08 marks]

(c). Figure Q2 shows a circuit diagram setup to charge a capacitor. Let the switch turn on at time $t = 0$. Draw the **charging current graph** and **charging voltage graph** of the capacitor and mark the **time constant** on the graphs.[03 marks]

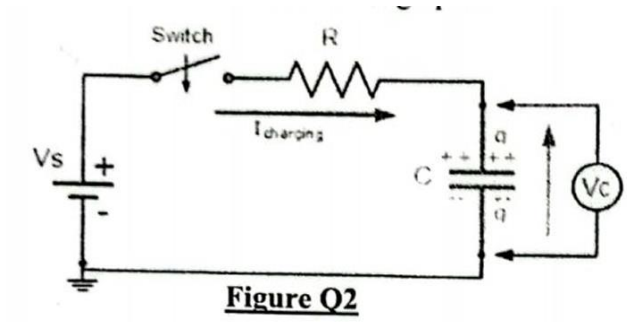

Figure Q2

(d). Consider the same circuit as in Question 2 (c) part. Capacitance C is $1\mu\mathrm{F}$, resistor R is $2\ \mathrm{M}\Omega$ and supply voltage V_S is $50\ \mathrm{V}$. Let the switch is closed at time $t = 0$. Calculate the followings: (i). The time constant. (ii). Initial charging current. (iii). The voltage across the capacitor in $5\ \mathrm{seconds}$ after closing the switch. (iv). The time taken for the capacitor to charge fully.[07 marks]

Question 3

(a). Briefly explain the purpose of **magnetic shielding** in electric circuits. Briefly explain the characteristics of shielding materials.[04 marks]

(b). State three factors that determine the **reluctance** of a magnetic circuit and also define the reluctance of a magnetic circuit.[04 marks]

(c). A magnetic core structure made of a **cast iron bar** is shown in figure Q3. The cross-sectional area of the bar is $200\ \mathrm{mm}^2$ and other dimensions are shown in the figure. Number of turns of the magnetizing coil in the right-most limb is 90. The relative permeability of the core material is 600. Determine the current that should be carried by the magnetizing coil to produce a $2\ \mathrm{mWb}$ of flux in the left-most limb. Flux leakage is negligible.[06 marks]

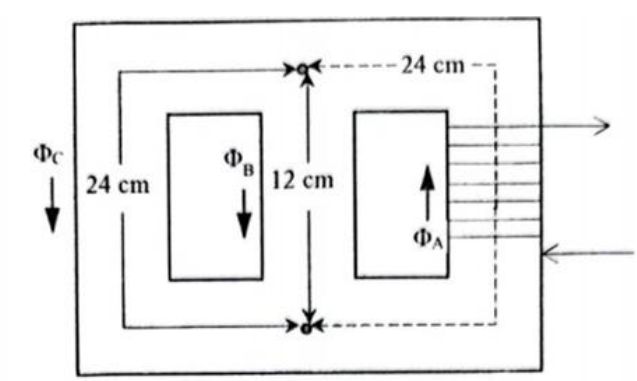

Figure Q3

(d). A coil of $250\ \text{turns}$, gives $12\ \text{mH}$ inductance when it is wound on a non-magnetic material. (i). If a current of $5\ \text{A}$ flows through the coil, determine the flux produced by it. (ii). If the current flow direction is reversed in $8\ \text{ms}$, determine the value of induced emf.[06 marks]

Question 4

(a). As humans, we cannot directly visualize the current flowing in a conductor. But the effects of the current flow are sensible. State **4 effects of electric current**.[04 marks]

(b). In the energy meter used by **Ceylon Electricity Board's (CEB) applications**, Watt-second or Joule is not used as a unit of measure of electrical energy. Explain the reason for that and mention the unit, used in CEB applications.[04 marks]

(c). Electrical equipment in a factory is connected to $240\ \text{V}$ main supply. When it is operated for $30\ \text{minutes}$, it consumes $5\ \text{MJ}$ of energy. (i). Find the **power rating** of the equipment. (ii). If the cost for a unit of electricity is $\text{Rs. } 75.00$, find the cost if the equipment is operated for $6\ \text{hours}$.[06 marks]

(d). A soldering iron is rated at $60\ \text{Watts}$. When it is connected to a $240\ \text{V}$ main supply it takes $4\ \text{minutes}$ to heat to the temperature of 180°C from 20°C. The heating element of the soldering iron is made up of copper, which has a specific heat capacity of $390\ \text{J/kg/}^\circ\text{C}$. Assuming that the soldering iron is to be 100% efficient, Find the followings: (i). Amount of **heat released** by the heating element. (ii). **Mass** of the heating element.[06 marks]

Question 5

(a). With related to ac circuits, briefly explain the following terms: (i). **Power factor**. (ii). **Active power** and **Reactive power**. [04 marks]

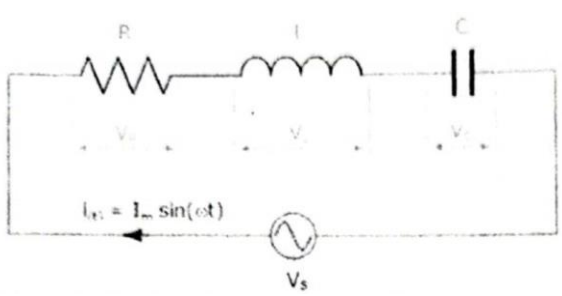

Figure Q5

(b). Figure Q5 shows an ac circuit where an RLC load is connected to a $200\ \text{V}$, $50\ \text{Hz}$ ac supply. The parameter values of the load are $R = 15\Omega$, $L = 0.3\ \text{H}$ and $C = 100\mu\text{F}$. Find the followings: (i). The inductive impedance, capacitive impedances, and the total impedance. (ii). The circuit current (magnitude and angle). (iii). The voltage across each element R, L, and C. (iv). Active power and reactive power. [16 marks]

Question 6

(a). State three advantages of a **three-phase power system** over a single-phase power system. [04 marks]

(b). Draw two, 3-phase supply systems in which one is **delta-connected** and the other is **star-connected**. Write down the relationships between line voltages to phase voltages and line currents to phase currents in each supply system. [04 marks]

(c). A three-phase **delta-connected** balance load consumes $5\ \mathrm{kW}$ per phase. The load in each phase is a coil of resistance 20Ω and inductance $191\ \mathrm{mH}$. The line current drawn by the load is $25.98\ \mathrm{A}$. Find the line voltage and phase voltage. [04 marks]

(d). A 3-phase **delta connected load** is connected to a main supply of $400\ \mathrm{V}$, $50\ \mathrm{Hz}$. The load comprises of a 20Ω resistor and a $90\mu\mathrm{F}$ capacitor in series in each phase. Find: (i). Line current and the phase current. (ii). Total power dissipation. (iii). kVA rating of the load. Draw the complete phasor diagram for the load. [08 marks]